



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY
Preliminary Examination
Paper 2

9647/02
19 August 2011
2 hours

Additional Materials: Data Booklet

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part questions.

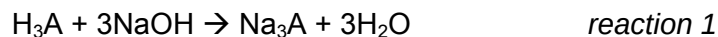
For Examiner's Use	
1	/12
2	/22
3	/10
4	/8
5	/10
6	/10
TOTAL	/72

This document consist of **15** printed pages and **1** blank page

1 Planning (P)

You are required to calculate the enthalpy change of neutralisation, $\Delta H_{\text{neutralisation}}$, for the reaction between solid citric acid, $\text{C(OH)(COOH)(CH}_2\text{COOH)}_2\cdot\text{H}_2\text{O}$ and aqueous sodium hydroxide.

Citric acid is a triprotic acid which can be represented as H_3A . The equation for the reaction is as follows:



FA 1 contains an aqueous solution of sodium hydroxide, NaOH , of unknown concentration. You may assume that the NaOH is in excess.

FA 2 contains 3.000 g of solid citric acid.

- (a) Write a plan for the determination of the enthalpy change of neutralisation between **FA 1** and **FA 2**. You may use the reagents and apparatus normally found in a school or college laboratory.

In your plan you should give:

the essential details of the calorimetric procedure;

all appropriate tables to record the readings;

safety precaution(s) that are employed to minimise the spillage of the chemicals during the experiment.

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..... [6]

- (b)** A student decides to modify the above experiment by using a solution of **FA 2** instead of solid **FA 2**, whilst keeping the total volume of solution mixture constant. Suggest how this modification would affect the experiment. Explain your answer.

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..... [1]

- (c) Using the values that you have proposed in (a), calculate the $\Delta H_{\text{neutralisation}}$ for *reaction 1* in terms of ΔT .
[4.3 J are required to raise the temperature of 1.0 cm³ of solution by 1.0 °C]

[3]

- (d) To obtain an anhydrous sample of solid citric acid, a student proposes the following procedure:

Weigh the empty crucible provided. Then weigh it again with solid **FA 2** of about 0.100 g.

At first, gently warm the solid in the crucible otherwise the solid will “spit” out of the crucible. Then increase the heat gradually. Heat for about 8 minutes. Allow to cool on a heat proof mat, then reweigh. Then heat the crucible with the contents strongly for about 2 minutes. Repeat the heating, cooling, reweighing procedure until two consecutive weightings are within 0.010 g of each other.

- (i) Suggest **one** potential problem that may arise should such a procedure be adopted and state how you would modify the procedure to minimise the problem.

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- (ii) It was noted by the student that the compound became black and charred after 8 minutes of heating. Suggest a reason why this might have occurred and state how you would modify the procedure to minimise the problem.

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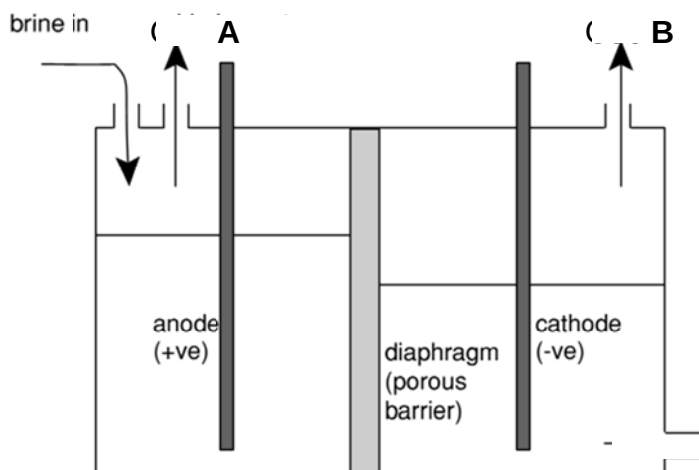
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[2]

[Total: 12]

2 This is a question on the halogens and their compounds.

- (a) Chlorine can be manufactured by the electrolysis of brine, concentrated NaCl, using a diaphragm cell. During the electrolysis process, a steady current of 3.0 A was passed through the cell. 500 cm³ of chlorine gas was collected at 30 °C and 1.5 atm.



- (i) Write ionic equations for the reactions occurring at the anode and at the cathode.

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- (ii) Determine the amount of chlorine produced, stating any assumptions that you have made.

- (iii) Hence, calculate the time taken for the process.

- (iv) If the process was carried out at 5 °C, will your assumption in (a)(ii) still be valid? Why?

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- (v) With reference to relevant electrode potentials in the *Data Booklet*, explain why fluorine gas cannot be prepared by electrolysis of aqueous sodium fluoride, NaF(aq).

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[11]

- (b) One major use of chlorine is in the bleaching process. When chlorine is reacted with aqueous sodium hydroxide at 20 °C, sodium chlorate(I), NaClO is formed.

- (i) Write a balanced equation for the reaction between chlorine and aqueous sodium hydroxide.

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- (ii) In the reaction in (b)(i), chlorine undergoes *disproportionation*. With reference to the changes in the oxidation states of chlorine, explain what is meant by the term in *italics*.

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- (iii) Similar disproportionation occurs for bromine and iodine at even lower temperatures with the following observations:

At 0 °C,

Bromine reacts readily with aqueous NaOH to form BrO⁻ (+1).

Iodine reacts readily with aqueous NaOH to form IO₃⁻ (+5).

Suggest the relative stability of higher oxidation states from Cl to I.

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[3]

- (c) Manganese ions, $\text{Mn}^{2+}(\text{aq})$, present in water supplies can cause discolouration in laundered goods and deposits on industrial machines. Though chlorine can be used to control these problems, it reacts so slowly that $\text{Mn}^{2+}(\text{aq})$ ions may still be present in the water distribution system after 24 hours. Chlorine dioxide reacts much more rapidly with $\text{Mn}^{2+}(\text{aq})$ in basic medium, oxidising it to manganese(IV) oxide. Chlorine dioxide is reduced to $\text{ClO}_2^- (\text{aq})$ ions.

- (i) Draw the dot-and-cross diagram for chlorine dioxide and suggest a reason why chlorine dioxide is a good oxidising agent.

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- (ii) With the aid of relevant half-equations, construct an equation for the overall reaction.

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[4]

- (d) Account for the following observations:

- (i) At around 30°C , the RMM of hydrogen fluoride appears to be 40.0 and above 60°C , it is about 20.0.

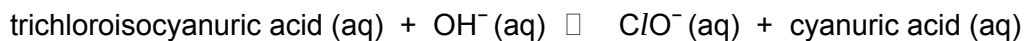
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- (ii) Sodium chloride reacts with concentrated sulfuric acid to form hydrogen chloride while sodium bromide reacts with concentrated sulfuric acid to form bromine.

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[2]

- (e) The use of chlorine as a disinfectant in swimming pools is now widely banned and the weak acid trichloroisocyanuric acid is used instead.



The $\text{ClO}^{\ominus}(\text{aq})$ is an effective disinfectant.

The graph below shows a decrease in the concentration of trichloroisocyanuric acid as the reaction progresses. At time t , some trichloroisocyanuric acid is added until the concentration is increased to 0.15 mol dm^{-3} . Sketch, on the same axes, how the concentration of trichloroisocyanuric acid changes from time t , till a new equilibrium is achieved.

[trichloroisocyanuric acid] / mol dm^{-3}



[2]

[Total: 22]

3(a) A hydrocarbon burns in excess oxygen forming 480 cm³ of carbon dioxide gas and 0.36 g of water at room temperature and pressure. The hydrocarbon reacts with chlorine dissolved in tetrachloromethane to form a compound with only 1 chiral centre.

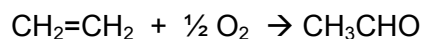
(i) Deduce the structure of a hydrocarbon containing four carbon atoms which fulfils the above requirements.

(ii) Name the type of reaction and describe the mechanism of the reaction between the hydrocarbon and chlorine.

(iii) The product from (a)(ii) is then heated with ethanolic sodium hydroxide. Three different structural isomers of monochloroalkenes are obtained. Draw the structures of **any two** monochloroalkenes.

[7]

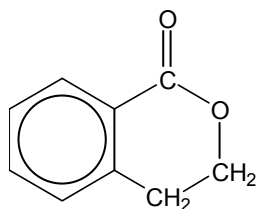
- (b) Aldehydes and ketones are produced industrially by the catalytic oxidation of alkenes, e.g. ethanal is manufactured from ethene as shown below.



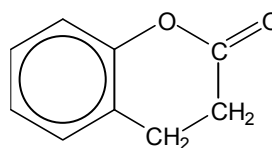
This process is also used industrially with but-2-ene. Draw the displayed formula of the product which would be produced from but-2-ene.

[1]

- (c) Organic compounds can be distinguished using simple chemical tests. Suggest how you can distinguish between the following pair of isomers. State clearly the reagents and conditions used as well as the observations for each compound.



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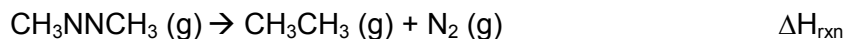
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[2]

[Total: 10]

- 4 The decomposition of azomethane, CH_3NNCH_3 , is a homogenous and unimolecular reaction. When heated, azomethane decomposes to ethane and nitrogen as shown in the equation below:



- (a) Predict if the change in entropy of the reaction is positive or negative. Justify your answer.

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[2]

- (b) The following data are given:

Standard enthalpy change of formation of ethane (g) = $-83.85 \text{ kJ mol}^{-1}$

Standard enthalpy change of formation of azomethane (g) = $148.8 \text{ kJ mol}^{-1}$

Use the given data to calculate the enthalpy change of reaction for the decomposition of azomethane.

[2]

- (c) A structural isomer of azomethane is $\text{NH}_2\text{CH}=\text{CHNH}_2$, identify the type of hybridisation about the C atom and explain why the π bond in $\text{NH}_2\text{CH}=\text{CHNH}_2$ is described as a localised π bond.

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[2]

- (d) Explain, in terms of structure and bonding, why CH_3CH_3 is more soluble in tetrachloromethane than $\text{NH}_2\text{CH}=\text{CHNH}_2$.

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[2]

[Total: 8]

5 Elements **E**, **F** and **G** are period 3 elements.

- (a) **E** is a white solid with melting point of 317 K. When exposed to air, it reacts spontaneously, producing a white powder of empirical formula E_2O_5 . E_2O_5 readily undergoes neutralisation with aqueous NaOH to form salt and water.

(i) From the observations above, deduce the identity of **E**.

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(ii) Based on your deduced identity of **E**, describe the above reactions using only equations with state symbols.

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(iii) With the aid of an equation, suggest and account for a pH value for the resultant solution when the oxide of **E** dissolves in water.

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[6]

- (b) Element **F** forms an **amphoteric** oxide readily in air. Using only equations with state symbols, explain what is meant by the word in bold.

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[2]

(c) The successive ionisation energies of element **G** is shown below.

	1st	2nd	3rd	4th	5th	6th	7 th	8 th
I.E. / kJmol ⁻¹	999	2265	3331	5071	7008	8487	27107	31719

(i) Deduce the group number in which **G** belongs.

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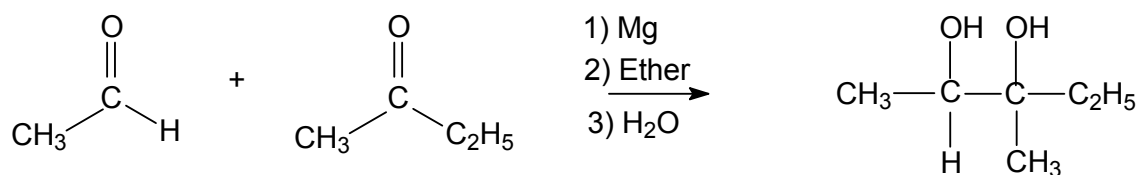
(ii) Write down the electronic configuration of **G** in the +4 oxidation state.

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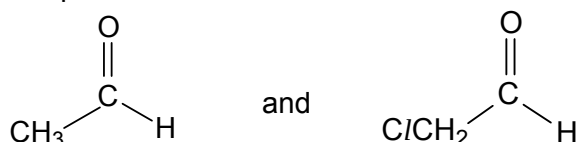
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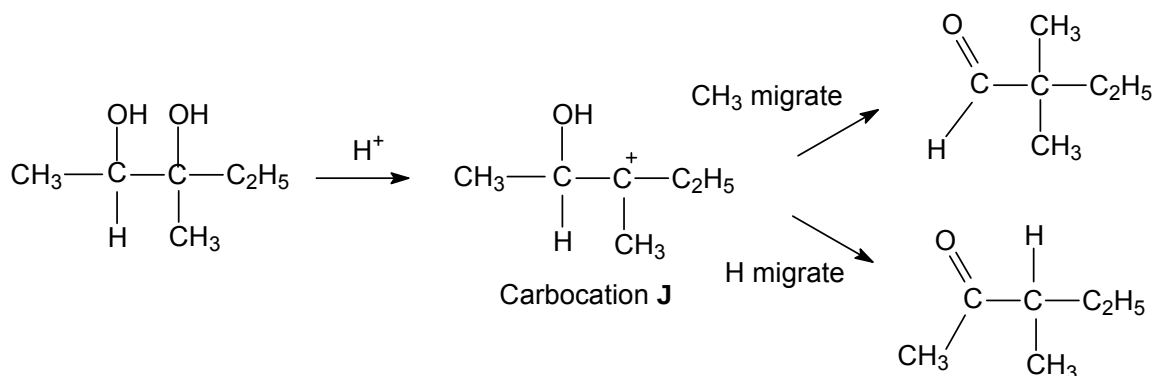
- 6 Two carbonyl groups can be coupled together to form a diol, as seen from the reaction below.



- (a) (i) Draw the displayed formula of the diol that will be formed when the following compounds are reacted.



The Pinacol rearrangement involves the formation of carbonyl compounds from a diol via the following reaction scheme.



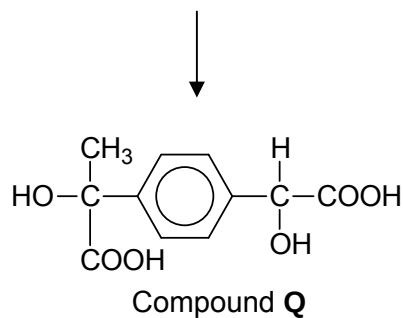
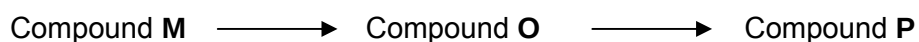
- (ii) Another carbocation **K**, other than **J**, can also be formed. Draw the structural formula of **K** and suggest a reason why the yield of **K** is smaller.
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- (iii) Draw the more stable carbonyl compounds formed when your diol in (a)(i) undergoes the Pinacol rearrangement.

[5]

- (b) A compound **L**, $C_9H_{10}Br_2$, undergoes reaction with aqueous sodium hydroxide under reflux to give compound **M**, $C_9H_{12}O_2$. Compound **M** then undergoes nucleophilic substitution with $ClCOCH_2COCl$ to form a cyclic compound **N**. Compound **M** gives a yellow precipitate when it undergoes oxidation with warm alkaline aqueous iodine.

Compound **M** can also undergo a series of reactions as shown below to yield Compound **Q**.



- (i) Draw the structural formulae of compounds **M**, **N**, **O** and **P**.

- (ii) State the reagents and conditions for converting compound **O** to **P**.

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[5]

[Total: 10]

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